
Advanced Statistical Physics

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Exercises: Stephan Kleinbölting

Sheet 10

Due date: 03.02.2021 12:00

To be discussed on: 08.02.2021

18 Finite-size scaling

6 pts.

It was demonstrated in the lectures how the renormalization group relates systems at different sizes, in particular to the system in the thermodynamic limit, thus explaining *finite-size scaling* behavior near criticality.

In this problem we return to the dataset assembled on sheet 4 and verify the scaling behavior.

Note: If your simulation code ran well before, use it to generate data for the system sizes indicated below. If not, you are free to use the sample data that was provided with the solution to sheet 4. Please make a note where your data comes from.

- (a) (3 pts) The finite-size scaling of the susceptibility $\chi(t, l^{-1}) \sim l^{\gamma/\nu} \Psi(l^{1/\nu} t)$ was already derived in the lecture.

Plot the appropriately rescaled susceptibilities for system sizes $L = 20, 30, 40, 50$ into one diagram. They should collapse onto a universal scaling function.

- (b) (3 pts) In the same vein, deduce the finite-size scaling behavior of the specific heat $c = \partial_T^2 f_s$ from the scaling relation of the free energy density¹

$$f_s(t, h, l^{-1}) = b^{-d} f_s(t', h', bl^{-1}) = |t|^{d/y_t} D_{\pm} \left(h t^{y_h/y_t}, l^{-1} t^{-1/y_t} \right). \quad (1)$$

Plot the appropriately rescaled heat capacities into one diagram.

Note: An exponent of zero indicates logarithmic scaling.

¹scale factors t_0 and h_0 are suppressed

19 Coupled scalar fields ²

14 pts.

Consider the following Hamiltonian

$$\beta H[m(\mathbf{x}), \phi(\mathbf{x})] = \int d\mathbf{x} \left\{ \frac{K}{2} [\nabla m(\mathbf{x})]^2 + \frac{t}{2} m^2(\mathbf{x}) - h m(\mathbf{x}) + \frac{L}{2} [\nabla^2 \phi(\mathbf{x})]^2 + v \nabla m(\mathbf{x}) \cdot \nabla \phi(\mathbf{x}) \right\}, \quad (2)$$

which describes two interacting real scalar fields $m(\mathbf{x})$ and $\phi(\mathbf{x})$. This Hamiltonian is defined with an implicit high-frequency cutoff Λ .

- (a) (4 pts) Write the Hamiltonian βH and the partition function Z in terms of the Fourier transforms of the scalar fields $m(\mathbf{q})$ and $\phi(\mathbf{q})$, where

$$m(\mathbf{q}) = \int d^d \mathbf{x} m(\mathbf{x}) e^{i\mathbf{q} \cdot \mathbf{x}} \quad (3)$$

$$m(\mathbf{x}) = \int \frac{d^d \mathbf{q}}{(2\pi)^d} m(\mathbf{q}) e^{-i\mathbf{q} \cdot \mathbf{x}} \quad (4)$$

and similarly for $\phi(\mathbf{q})$ and $\phi(\mathbf{x})$.

- (b) (2 pts) In order to analyze the behaviour of the couplings t, K, h, L, v under a renormalisation step, we start by dividing the Fourier modes into slow modes ($|\mathbf{q}| \leq \Lambda/b$) and fast modes ($\Lambda/b < |\mathbf{q}| \leq \Lambda$). You can make the distinction between fast and slow Fourier modes explicit by defining $m(\mathbf{q}) \equiv m_s(\mathbf{q}) + m_f(\mathbf{q})$ (and similarly for $\phi(\mathbf{q})$), where

$$m_s(\mathbf{q}) = \begin{cases} m(\mathbf{q}) & \text{for } 0 \leq |\mathbf{q}| < \Lambda/b \\ 0 & \text{for } \Lambda/b \leq |\mathbf{q}| \leq \Lambda \end{cases} \quad (5)$$

$$m_f(\mathbf{q}) = \begin{cases} 0 & \text{for } 0 \leq |\mathbf{q}| < \Lambda/b \\ m(\mathbf{q}) & \text{for } \Lambda/b \leq |\mathbf{q}| \leq \Lambda \end{cases} \quad (6)$$

We will then ‘integrate out the fast modes’ by integrating over the $m_f(\mathbf{q})$ and $\phi_f(\mathbf{q})$ in the partition function Z . Do not evaluate integrals that only contribute a constant additive term (stemming from the fast Fourier modes).

- (c) (3 pts) Rescale the integral over the slow Fourier modes by making the transformation $\mathbf{q}' = b\mathbf{q}$. Rescale the fields such that $m'(\mathbf{q}') = m_s(\mathbf{q})/z_m$ and $\phi'(\mathbf{q}') = \phi_s(\mathbf{q})/z_\phi$. Express the new parameters $\{K', t', h', L', v'\}$ in terms of the original parameters $\{K, t, h, L, v\}$ involving functions of the variables $\{b, d, z_m, z_\phi\}$.
- (d) (3 pts) z_m and z_ϕ are scaling parameters which we can choose for convenience (they affect the amplitude of the new fields m' and ϕ'). Find a choice of these parameters that leaves the couplings K and L unchanged³. Consider the renormalization transformation associated to the scaling parameters $z_m = b^{\frac{d}{2}+1}$ and $z_\phi = b^{\frac{d}{2}+2}$. Find the corresponding fixed point and compute the renormalization groups eigenvalues y_t, y_h, y_v .
- (e) (2 pts) Close to this fixed point, the singular part of the free energy has a scaling form $f(t, h, v) = t^{2-\alpha} g(h/t^\Delta, v/t^\omega)$. Use the results of (d) to find $\{\alpha, \Delta, \omega\}$.

To do that, recall that the singular parts of the free energies of the system before and after the renormalization step are related by (why?)

$$f(t, h, v) = b^{-d} f(t', h', v'), \quad (7)$$

then choose a scaling factor b such that $b = t^{-1/y_t}$.

²with a hat tip to Mehran Kardar

³You can compare your results to $z_m = b^{\frac{d}{2}+1}$ and $z_\phi = b^{\frac{d}{2}+2}$